

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] JULY-2023

Paper Code: AIDS/AIML/IOT-204

Subject: Database Management Systems

Time : 3 Hours

Maximum Marks : 75

Note: Attempt five questions in all including question no. 1 which is compulsory. Select one question from each unit. Scientific calculators are allowed.

Q1 Answer the following questions:

(2.5×6=15)

- (a) Distributed vs Object-oriented databases
- (b) Explain ACID property in brief
- (c) Discuss three tier architecture of DBMS with example
- (d) Explain the concept of referential integrity giving example
- (e) Write short note on key features of MongoDB
- (f) Tuple vs Domain Relational Calculus

UNIT-I

- Q2 (a) A database is being built up by an e-commerce corporation "ABC" to track vendors and their products. This necessitates writing down the details such as name, mailing address, contact no. and email for each seller, the name, price, and quantity of each product, information like which seller's merchandise it is, a number that is specific to each product. Choose additional attributes for the schema that seems appropriate. Create an entity-relationship (ER) diagram to visualise this data. Be sure to note the restrictions on the relevant relationships, and give the entities the proper primary keys. (10)
- (b) Explain the concept of weak and strong entity set with the help of an example. Draw the ER diagram to represent these entities. (5)

OR

- Q3 (a) Write short note on the following with example: (7)
- i. Multivalued and Derived attributes
 - ii. Specialization vs Generalization
- (b) Explain how the hierarchical, network, and relational DBMS models differ from one another. Any real-time application of each should be given. (8)

UNIT-II

- Q4 (a) Explain various steps of query processing with the help of diagram. (7)
- (b) Explain the concept of join operation in DBMS relations. Differentiate between outer and inner join operations with examples. (8)

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P-1/2

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OR

- Q5 (a) Write short note on the following with example: (7)
 i. Query optimization techniques
 ii. Types of Cursors in PL/SQL
- (b) Explain the concept Nested queries in SQL. Write syntax of given clauses in SQL such as: **GROUP BY, DISTINCT, AVG(), LIKE, AND, CONCAT()**, with the help of examples. (2+6)

UNIT-III

- Q6 (a) What is concurrency? Discuss various concurrency problems associated with it. Explain Time-stamp based protocol for concurrency control. (8)
- (b) Let there are three concurrent transactions T1, T2 and T3. Determine the following schedule 'S' is conflict serializable or not? For a serializable schedule, find equivalent serial schedule. (7)
S: R3(A), R2(A), W3(A), R1(A), W1(A)

OR

- Q7 (a) Define BCNF. Compare 3NF and BCNF. Which normal form is stronger between these two? Give an example of a relation in 3NF that is not in BCNF. (8)
- (b) Write short note **any two** from the following with example: (7)
 i. Functional dependency
 ii. Two phase locking protocol
 iii. View Serializability

UNIT-IV

- Q8 (a) Explain the architectures of distributed databases. How data is fragmented in distributed databases? Also, discuss how transparency is achieved in such databases. (8)
- (b) Discuss the need of data security in DBMS. Explain various security measures to protect database against various security threats. (7)

OR

- Q9 (a) Design a data model for an e-commerce website that stores information on users, products, and orders. Evaluate the strengths and weaknesses of an RDBMS versus a NoSQL database for this application. (8)
- (b) Write short note on the following with example: (7)
 i. Applications of Data mining
 ii. CAP theorem
